A four-dimensional periodic Costas array of order six

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Abstract

We give a bijection from $[2]\times[3]$ onto itself whose graph, extended periodically in four coordinates, has distinct nonzero ordered ordinary difference vectors in every $2\times3\times2\times3$ window. Its order is 6, whereas the proposed minimum-order restriction of Rubio and Torres would require 4. A short proof keeps the signs of the two binary-coordinate differences and reduces the other two differences modulo three.

1 The periodicity question

Write $[r]=\{1,\dots,r\}$. Rubio and Torres [1] define a multidimensional permutation array through a bijection

$$\varphi:\prod_{i=1}^{k}[n_i]\longrightarrow\prod_{i=k+1}^{m}[n_i],\qquad n=\prod_{i=1}^{k}n_i=\prod_{i=k+1}^{m}n_i,$$

where $m\geq2$ and all side lengths $n_i\geq2$. If S is its graph and $L=n_1\mathbb{Z}\times\cdots\times n_m\mathbb{Z}$, the array is periodic Costas when, for every $t\in\mathbb{Z}^m$, the nonzero ordered differences between points of

$$(S+L)\cap\left(t+\prod_{i=1}^{m}[n_i]\right)$$

are distinct. These differences are ordinary integer vectors. In Conjecture 1 of [1], Rubio and Torres propose that, when $1\leq m-k\leq k$, periodicity forces $n=2^k$, or equivalently $n_1=\cdots=n_k=2$.

The six-point array below contradicts this restriction in dimension four. An earlier public Prove2Me record by shivm, dated 4 October 2026, labels the same conjecture disproved [2]. We have not inspected the proof or witness underlying that record. We acknowledge the earlier announcement and make no novelty or priority claim. The argument here is self-contained.

2 An array of order six

Let $\varphi : [2]\times[3] \to [2]\times[3]$ be the map in Table 1, and let $P_1, \dots, P_6$ denote the corresponding graph points.

x	$\varphi(x)$	$(x, \varphi(x))$
$(1,1)$	$(1,1)$	$P_1 = (1,1,1,1)$
$(1,2)$	$(1,2)$	$P_2 = (1,2,1,2)$
$(1,3)$	$(2,1)$	$P_3 = (1,3,2,1)$
$(2,1)$	$(1,3)$	$P_4 = (2,1,1,3)$
$(2,2)$	$(2,3)$	$P_5 = (2,2,2,3)$
$(2,3)$	$(2,2)$	$P_6 = (2,3,2,2)$

Table 1: The bijection defining the array.

Theorem 1. *Set $S = \{P_1, \dots, P_6\}$ and $L = 2\mathbb{Z} \times 3\mathbb{Z} \times 2\mathbb{Z} \times 3\mathbb{Z}$. For every $t \in \mathbb{Z}^4$, all nonzero ordered ordinary differences between points of*

$$S_t = (S + L) \cap (t + [2] \times [3] \times [2] \times [3])$$

are distinct. Thus S is a periodic four-dimensional Costas array of order 6 with $k = 2$, and $6 > 2^2$.

Proof. The input and output columns of Table 1 each enumerate $[2]\times[3]$, so φ is a bijection. To distinguish its ordered displacements, retain the two binary-coordinate signs. For $P = (a, b, c, d)$ and $Q = (a', b', c', d')$ in S, define

$$H(P, Q) = \big(a' - a, \operatorname{res}_3(b' - b), c' - c, \operatorname{res}_3(d' - d)\big), \tag{1}$$

where $\operatorname{res}_N(z)$ is the unique remainder in $\{0, \dots, N-1\}$ of the integer z modulo N.

Table 2 gives the residue pairs in coordinates two and four, grouped by the ordinary differences in coordinates one and three. The ordered pair (i, j) in the middle column means (P_i, P_j).

$(a' - a, c' - c)$	Ordered pairs	$(\operatorname{res}_3(b' - b), \operatorname{res}_3(d' - d))$
$(0,0)$	$(1,2), (2,1), (5,6), (6,5)$	$(1,1), (2,2), (1,2), (2,1)$
$(0,1)$	$(1,3), (2,3), (4,5), (4,6)$	$(2,0), (1,2), (1,0), (2,2)$
$(1,0)$	$(1,4), (2,4), (3,5), (3,6)$	$(0,2), (2,1), (2,2), (0,1)$
$(1,1)$	$(1,5), (1,6), (2,5), (2,6)$	$(1,2), (2,1), (0,1), (1,0)$
$(1,-1)$	$(3,4)$	$(1,2)$

Table 2: Distinct residues within each fixed pair of binary differences.

Reversing the pairs in the last four rows gives the remaining binary differences

$$(0,-1), \quad (-1,0), \quad (-1,-1), \quad (-1,1)$$

and negates the two residues modulo three. Each row has distinct residue pairs, and different rows have different binary differences. The displayed pairs and these reversals are distinct and number $4 + 2(4+4+4+1) = 30 = 6(6-1)$, so they exhaust all ordered pairs of distinct points of S. Consequently H is injective on those pairs.

Now fix an arbitrary $t \in \mathbb{Z}^4$ and write $(n_1, n_2, n_3, n_4) = (2, 3, 2, 3)$. The representative of $u \in [n_i]$ in the interval $t_i + [n_i]$ is

$$F_{i,t}(u) = t_i + 1 + \mathrm{res}_{n_i}(u - t_i - 1).$$

It is congruent to u modulo n_i and lies in the stated interval. Two congruent integers in that interval differ by a multiple of n_i of absolute value at most $n_i - 1$, so they are equal. It follows that

$$P \longmapsto P_t = (F_{1,t}(p_1), \ldots, F_{4,t}(p_4))$$

is a bijection from S onto S_t.

For $i = 1, 3$, the values $F_{i,t}(1)$ and $F_{i,t}(2)$ are the two consecutive integers in $t_i + [2]$. Hence there is a fixed $\varepsilon_i \in \{-1, 1\}$, depending only on t_i, such that

$$F_{i,t}(u') - F_{i,t}(u) = \varepsilon_i(u' - u), \qquad u, u' \in [2].$$

For $i = 2, 4$, subtraction of congruences gives

$$\mathrm{res}_3(F_{i,t}(u') - F_{i,t}(u)) = \mathrm{res}_3(u' - u).$$

Suppose $Q_t - P_t = V_t - U_t$ as ordinary integer vectors, where $P, Q, U, V \in S$, $P \neq Q$ and $U \neq V$. The same signs $\varepsilon_1, \varepsilon_3$ occur for both pairs. Cancelling them in coordinates one and three, and reducing coordinates two and four modulo three, gives $H(P, Q) = H(U, V)$. Its injectivity implies $P = U$ and $Q = V$, and therefore $P_t = U_t$ and $Q_t = V_t$. Since t was arbitrary, the required distinct-difference property holds in every translated window. □

All four side lengths are at least two, $m - k = k = 2$, and both products of side lengths are 6. Thus Theorem 1 satisfies every premise of the proposed restriction while its conclusion fails.

AI use. OpenAI Codex assisted in finding the six-point example, developing the proof, drafting this note, and performing auxiliary exact-integer checks. Separate fresh-context Codex passes assessed readability and challenged the manuscript. Independent mathematical review is requested.

References

[1] I. Rubio and J. Torres, *Multidimensional Costas Arrays and Their Periodicity*, arXiv:2208.02378v1 (2022). Definitions 2–3 (p. 6), Definition 4 (p. 8), and Conjecture 1 (p. 16).

[2] shivm, *Rubio–Torres Conjecture 1: periodic Costas arrays have order* 2^k, Prove2Me, record dated 4 October 2026, marked "Disproved". https://prove2.me/theorems/f2732909-8aec-45d0-afe0-8943f5d2c80e. Accessed 15 October 2026; cited for the announcement.